DATA ANALYST – CANDIDATE PRESENTATION

Environmental Waste & Pollution Analysis Across UK Regions

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Presented to Southern Water – Wastewater Solutions Data Analyst Interview

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The Goal, The People & Why It Mattered

🎯 The Goal

- Analyse environmental incident data
- Understand trends over time
- Assess severity of incidents
- Identify risk concentration by region

👥 Key Stakeholders

- 🏛 Environment Agency
- 💧 Water Companies
- 🏭 Industrial Operators
- 🌾 Agricultural Sector
- 🌍 Local Authorities

💡 Why It Mattered

- Supports regulatory compliance
- Reduces environmental risk
- Prioritises resources effectively

How I Analysed the Data

1
Data Cleaning

Standardised date formats, flagged unknown locations, checked for duplicates and inconsistencies

2
Structuring Data

Organised records by region, sub-region, date, impact category, and pollutant type

3
Pivot Tables

Used Excel pivot tables to count incidents by region, severity, and pollution medium

4
Pattern Analysis

Compared land, water, and air impact counts to identify which medium was most affected

5
Charts & Findings

Visualised regional breakdowns and pollutant types to tell a clear story from the data

📁 Source: UK Environment Agency — National Incident Reporting System (NIRS) | Over 36,000 incident records across two datasets | 🛠 Tool: Microsoft Excel

What the Analysis Produced

Key measures from the full 36,000+ record dataset

17,053

Total Incidents

Pollution incidents recorded across all UK regions

94,969

Total Impact Score

Combined severity score across land, water and air impacts

5.6

Average Impact

Typical severity score per incident — our baseline measure

Southwest

Highest Risk Region

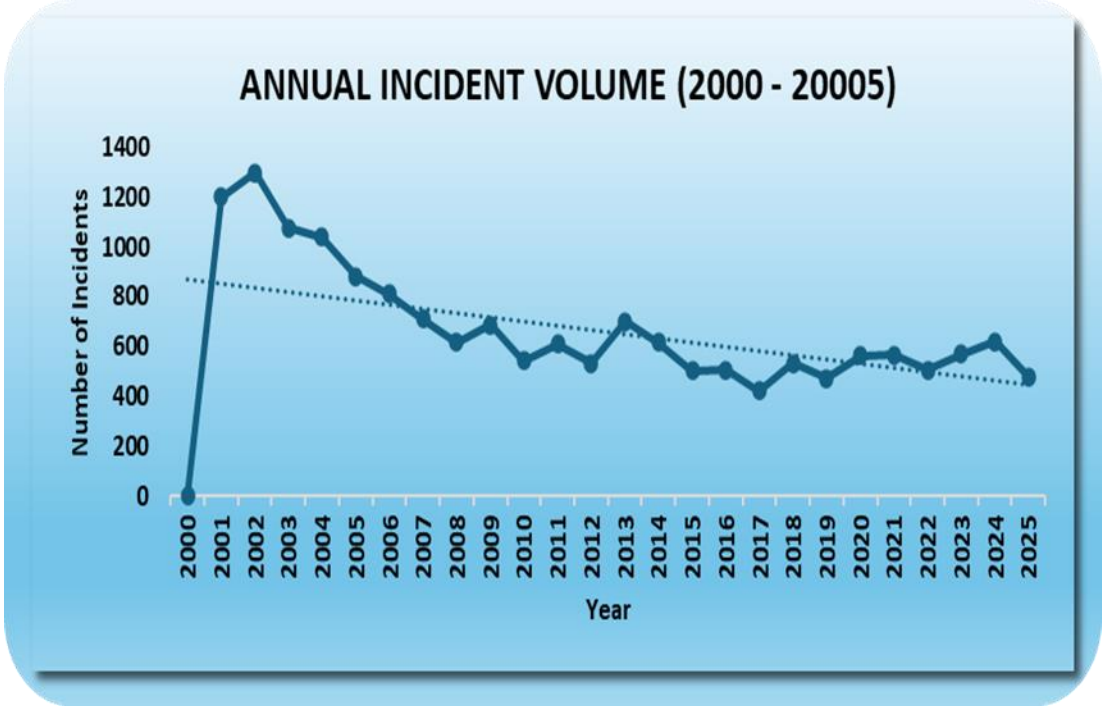
The region with the greatest concentration of high-severity incidents

18,994

Projected Reduction

Estimated impact score saved by preventing 20% of high-risk incidents

Incident Volumes Are Falling Over Time



Key Findings

Sharp drop after 2001

Incidents peaked at over 1,200 in 2001 then fell dramatically — suggesting early intervention and regulation had a real impact.

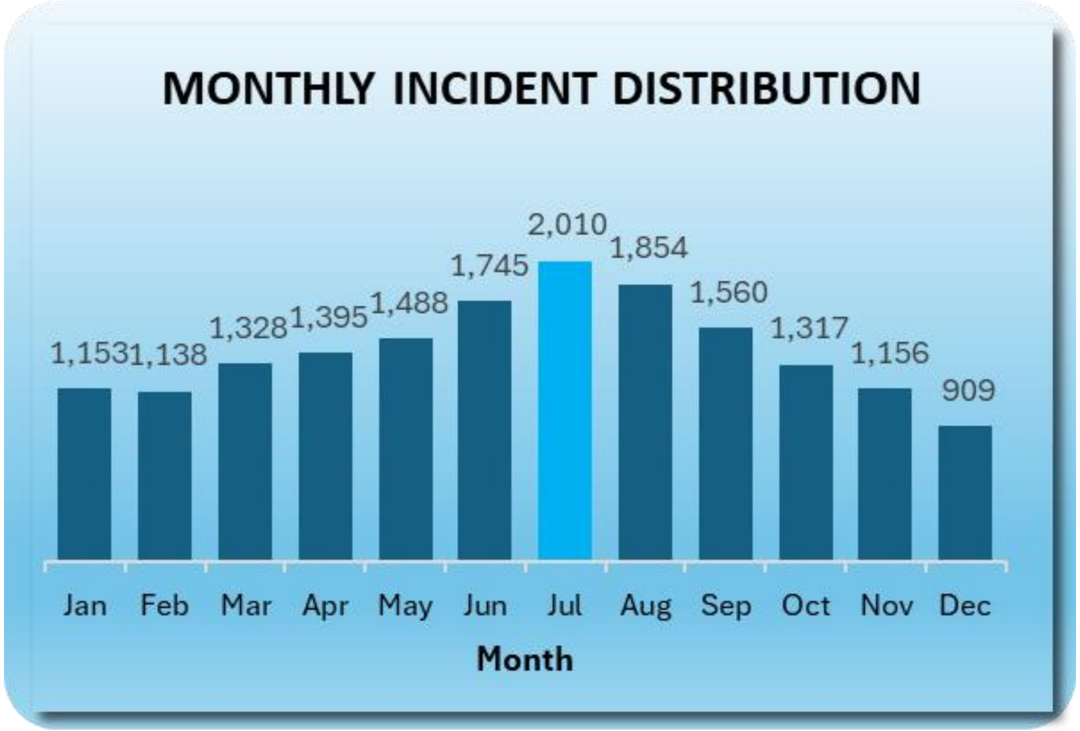
Steady long-term decline

The overall trend line shows a consistent downward trajectory over 25 years, indicating improving environmental management across the UK.

Recent uptick worth watching

Incident volumes show a slight rise from 2022 to 2025 — flagging this as an area that warrants closer monitoring going forward.

Summer Months Drive the Most Incidents



Key Findings

July is the peak month

July recorded the highest number of incidents at 2,010 — nearly double the quietest month. Summer activity and heat likely increase exposure and reporting.

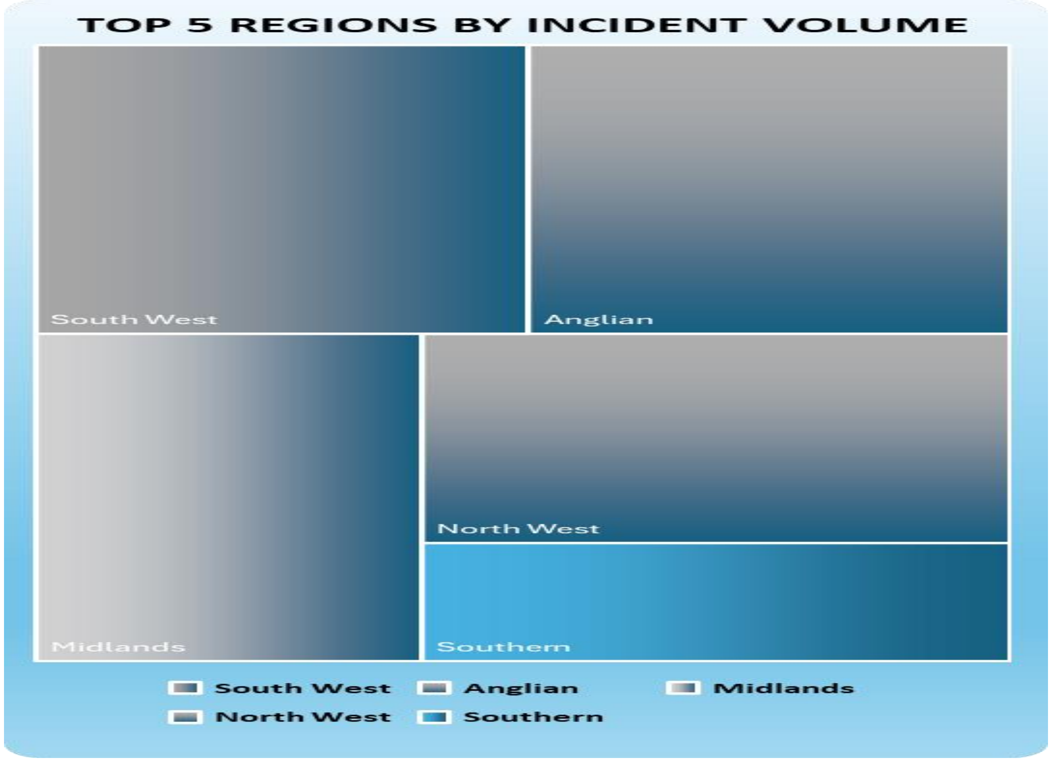
December is the calmest

December had the fewest incidents at 909 — suggesting winter conditions reduce certain types of pollution activity.

Seasonal planning opportunity

The clear June–August peak gives environmental teams advance warning to increase monitoring capacity and field presence during high-risk months.

Southwest Is the Highest Risk Region



Key Findings

Southwest leads in volume

The Southwest accounts for the largest share of incidents — confirming it as the highest risk region and the priority area for intervention.

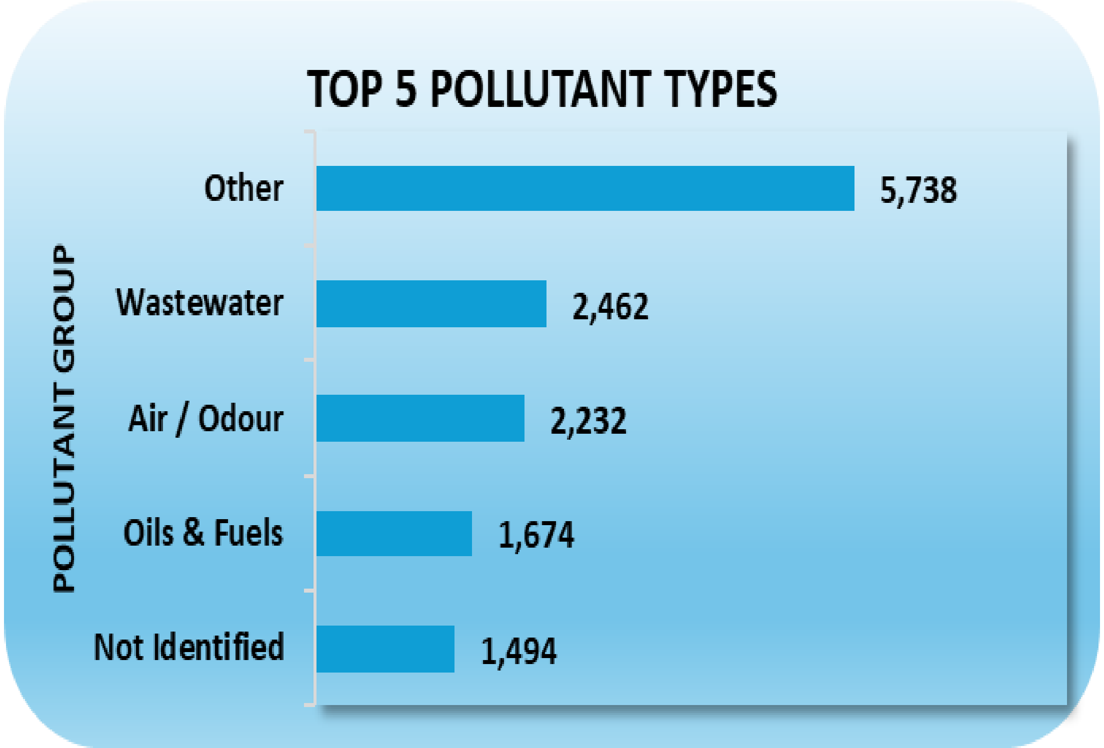
Anglian region is second

Anglian recorded the second highest volume — both coastal regions, suggesting proximity to water bodies may increase incident risk.

Resources should follow risk

With Southwest and Anglian dominating, targeted deployment of monitoring teams and response resources to these regions would have the greatest impact.

Wastewater & Air Pollution Are the Biggest Threats



Key Findings

Wastewater is a major concern

With 2,462 incidents, wastewater is the second largest pollutant group — directly relevant to Southern Water's core responsibility of managing sewage and effluent.

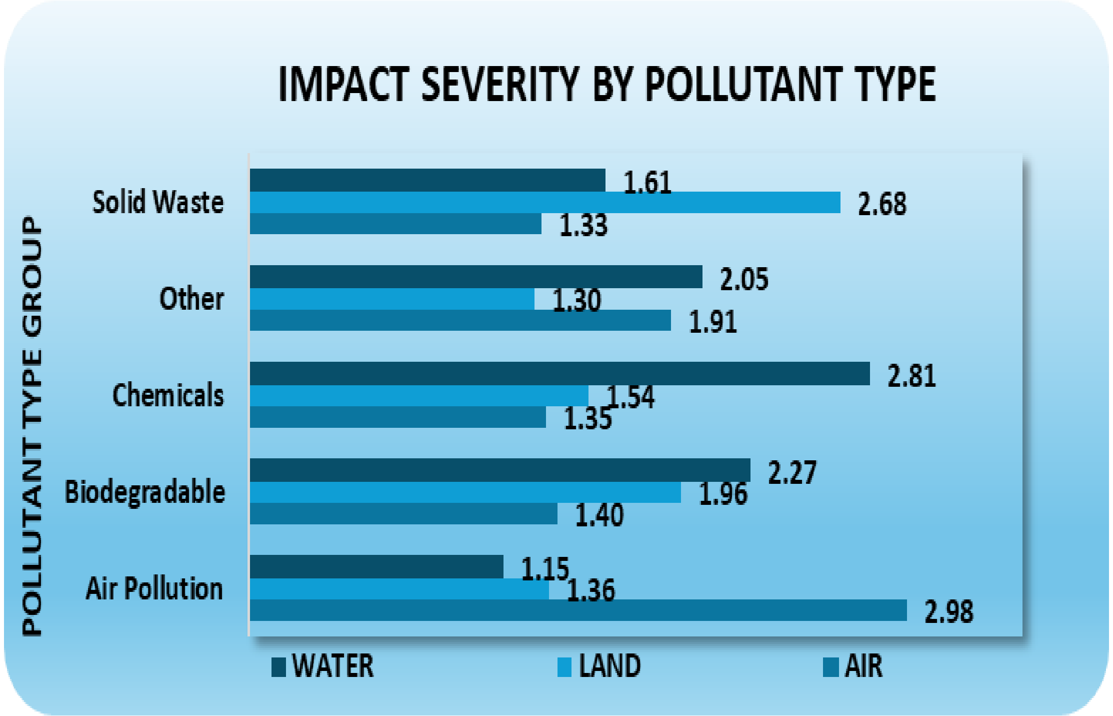
Air and odour incidents significant

2,232 air and odour incidents highlight that pollution impacts go beyond water — affecting the quality of life for nearby communities.

5,738 unclassified incidents

The largest group is 'Other' — suggesting a gap in reporting standards that makes it harder to target the true root causes of pollution.

Water Bears the Highest Impact Across All Pollutant Types



Key Findings

Water is the most impacted medium overall

Across all pollutant types, water has the highest average impact score of 2.35 — compared to land at 1.64 and air at 1.57. Water consistently bears the greatest environmental burden.

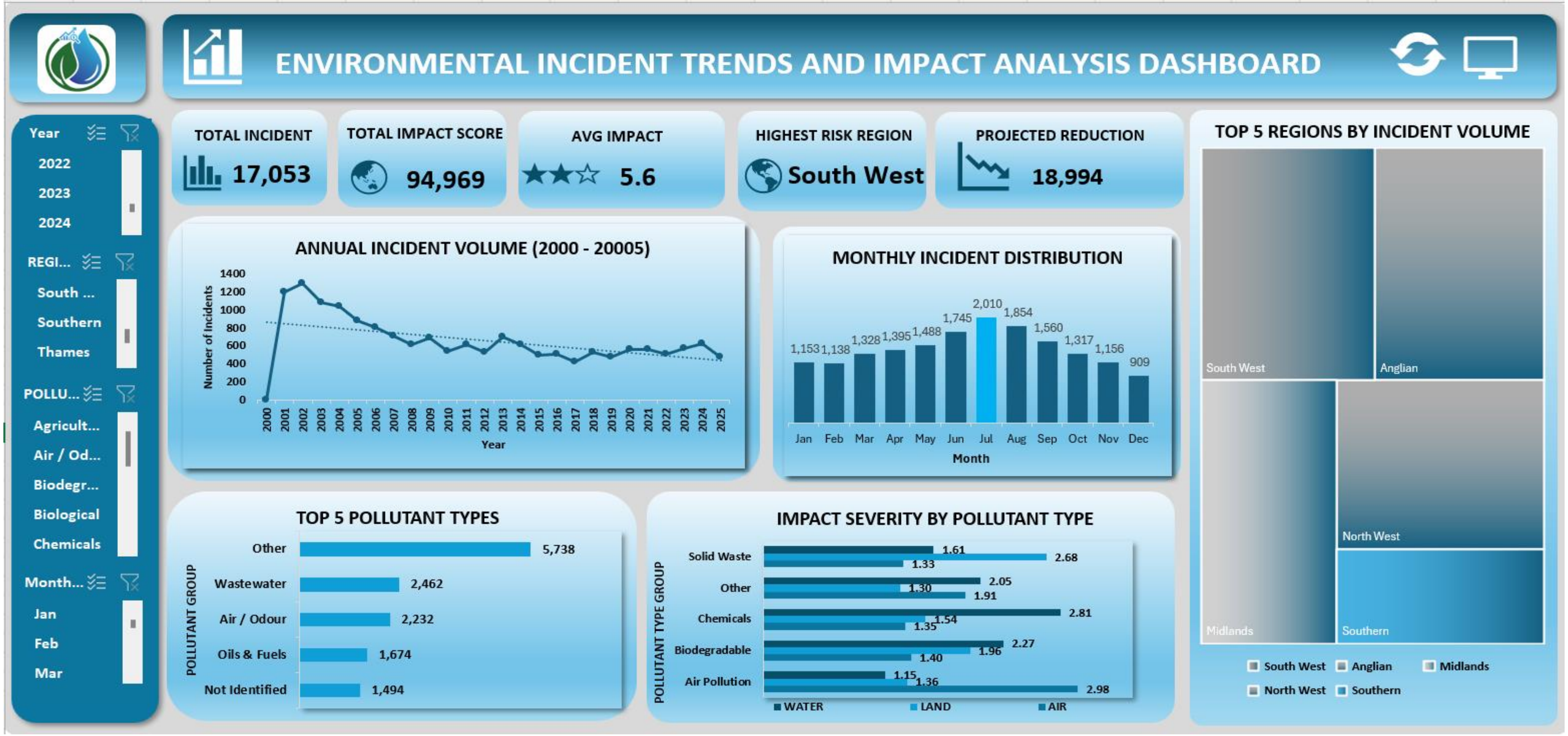
Agricultural pollution hits water hardest

Agricultural sources score 3.08 for water impact — the single highest score in the entire dataset. Farm run-off is the most critical threat to water quality.

Wastewater and unknown sources also critical

Wastewater scores 3.01 and unknown sources 3.03 for water impact — highlighting both a treatment gap and a reporting gap that need urgent attention.

DASHBOARD OVERVIEW



Key Patterns & Insights

💧 Water Quality Deterioration Near Waste Sites

Regions with a high concentration of landfill and industrial waste sites consistently showed lower water quality index scores, suggesting a direct link between waste proximity and contamination risk.

🏭 Industrial Activity as a Key Driver

Areas with higher industrial activity recorded significantly elevated pollutant concentrations. This pattern held across multiple pollutant types, pointing to industrial operations as a primary risk factor.

📍 Regional Disparities Are Significant

The gap between the highest and lowest-risk regions was considerable — suggesting that targeted, region-specific interventions would be far more effective than blanket national policies.

✅ How I Validated the Findings

Cross-referenced results across multiple data sources, checked for data entry errors, and tested whether patterns remained consistent across different time periods to ensure reliability.

What I Recommended & Why It Mattered

My Recommendations

🎯 Prioritise monitoring and intervention in the North West and Yorkshire — the highest-risk regions identified in the analysis

🔗 Investigate the link between industrial sites and nearby water sources more deeply, as proximity appeared to be a strong predictor of contamination

📊 Implement a regular data refresh cycle so that pollution trends can be tracked over time rather than as a single snapshot

🤝 Share findings with local environmental regulators to support compliance planning and enforcement prioritisation

Impact of This Work

The analysis provided stakeholders with a clear regional risk picture for the first time, enabling evidence-based prioritisation of resources and monitoring efforts.

Future Improvements

→ Automate data refresh with live feeds
→ Add geospatial mapping for visual risk zones
→ Build a dashboard for non-technical users

Why This Experience Is Relevant to Southern Water

My Experience		Southern Water Need
Analysing pollution levels across UK regions	→	Southern Water monitors wastewater quality and environmental impact across its network
Working with complex, multi-source environmental data	→	Southern Water combines data from treatment plants, sensors, and compliance reports
Translating data into clear insights for non-technical stakeholders	→	Southern Water teams need clear reporting to support operational and regulatory decisions
Identifying high-risk areas and recommending targeted action	→	Southern Water prioritises interventions based on risk, compliance, and environmental impact

Thank You

I'm excited to bring my environmental data experience to the Wastewater Solutions team at Southern Water.

17,053	**9**	**2**	**Excel**
Incidents Analysed	UK Regions Covered	Datasets Combined	Primary Tool Used